

PRODUCTIVITY

PRODUCTIVITY GROWTH

- Productivity growth is the increase in productivity from one period to the next relative to the productivity in the preceding period.

Productivity Growth =

$$\frac{\text{Current period productivity} - \text{Previous period productivity} \times 100}{\text{Previous period productivity}}$$

For example if productivity increased from 80 to 84, the growth rate would be -

$$\frac{84 - 80}{80} \times 100\% = 5\%$$

EXAMPLE

- A wrapping paper company produced 2000 rolls of paper one day. Standard price is 1Tk/roll. Labor cost was Tk. 160, material cost was Tk. 50 and overhead was Tk. 320. Determine the multifactor productivity.

- **Solution:**

$$\text{Multifactor productivity} = \frac{\text{Quantity produced @ standard price}}{\text{Labor cost} + \text{Material cost} + \text{Overhead}}$$

$$= \frac{2000 \text{ rolls} \times 1\text{Tk. / roll}}{160 + 50 + 320}$$

$$= 3.77 \text{ Tk output per Tk. input}$$

EXAMPLE

- Compute the multifactor productivity measure for each of the weeks shown. Assume 40 hrs weeks and an hourly wage Tk. 12. Overhead is 1.2 times to total weekly labor cost. Material cost is Tk. 3 per kg. Standard price is Tk. 140 per unit.

Week	Output (Units)	Workers	Materials (Kg)
1	300	6	45
2	338	7	46
3	322	7	46

- Also compute the productivity growth of week 2, 3 over week 1 of this problem.

EXAMPLE-SOLUTION

MFP (Week 1)

= Total Output /Multiple Inputs

= (300 units X TK 140 per unit)/ [(6 workers X 40 hrs per week X TK 12 per labor hour) + 1.2 (6 workers X 40 hrs per week X TK 12 per labor hour) + (45 Kg X TK 3 per Kg)]

= TK. 42000 / [(Tk. 2880) + 1.2 (TK. 2880) + (TK. 135)]

= TK. 42000 / [(Tk. 2880) + (TK. 3456) + (TK. 135)]

= TK. 42000 / TK. 6471

= 6.49 Tk output per Tk. input

MFP (Week 2)

= 6.28 Tk output per Tk. input

Productivity Growth of Week 2 =

[(Current period productivity – Previous period productivity) / Previous period productivity] x 100 %

= [(6.28-6.49) / 6.49] X 100%

= - 3.236%